

Conjecture 00000004082 is false:
the Kuramoto critical coupling is not a multiple of the mean
absolute frequency deviation

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1 The conjecture

Definition: Kuramoto synchronization is the phase locking of coupled oscillators, and the critical coupling K_c is the onset threshold of synchronization. *Conjecture:* For n all-to-all coupled oscillators, K_c depends only on the dispersion spectrum of the intrinsic frequencies, with formula $K_c = c_1 \cdot (\sum_i |\omega_i - \bar{\omega}|)/n$ for a universal constant c_1 .

The statement is a conjunction. We refute its formula clause, $K_c = c_1 \sum_i |\omega_i - \bar{\omega}|/n$, and in fact show that K_c is not *any* function of $\sum_i |\omega_i - \bar{\omega}|/n$. We do not use the undefined phrase “dispersion spectrum”.

2 Definitions (as in the Lean file)

Let $n \geq 1$, $\omega \in \mathbb{R}^n$ (intrinsic frequencies) and $K \in \mathbb{R}$. The all-to-all Kuramoto model is

$$\dot{\theta}_i = \omega_i + \frac{K}{n} \sum_{j=1}^n \sin(\theta_j - \theta_i), \quad i = 1, \dots, n.$$

Definition 1. ω is *locked at K* (**Locked**) if there are $\Omega \in \mathbb{R}$ and $\theta \in \mathbb{R}^n$ with $\omega_i + \frac{K}{n} \sum_j \sin(\theta_j - \theta_i) = \Omega$ for all i . A trajectory $\theta : \mathbb{R} \rightarrow \mathbb{R}^n$ is a solution (**IsKuramotoSolution**) if it has the derivative above at every time, and it is *phase locked* (**IsPhaseLocked**) if every difference $\theta_i(t) - \theta_j(t)$ is constant in t . The critical coupling is $K_c(\omega) = \inf\{K \geq 0 : \omega \text{ is locked at } K\}$ (**Kc**, a real **sInf**). We write $\bar{\omega} = \frac{1}{n} \sum_i \omega_i$ (**mean**) and $D(\omega) = \frac{1}{n} \sum_i |\omega_i - \bar{\omega}|$ (**meanAbsDev**).

Lemma 2 (**locked_iff_phaseLockedSolution**). ω is locked at K if and only if the Kuramoto ODE has a phase-locked solution.

Proof. If the equations hold, $\theta_i(t) = \theta_i + \Omega t$ is a solution, since the phase differences, and hence the right-hand side, do not depend on t . Conversely, if the differences are constant, then the derivatives at $t = 0$ of $\theta_i - \theta_j$ vanish, so all right-hand sides at $t = 0$ are equal to one number Ω . $\square$

3 Proof

Lemma 3 (**abs_sub_mean_le_of_locked**). If $n \geq 1$, $K \geq 0$ and ω is locked at K , then $|\omega_i - \bar{\omega}| \leq K$ for every i .

Proof. The double sum $\sum_i \sum_j \sin(\theta_j - \theta_i)$ is zero, because exchanging i and j changes its sign (**double_sum_sin_eq_zero**). Summing the n equations gives $\sum_i \omega_i = n\Omega$, so $\Omega = \bar{\omega}$. Then $|\omega_i - \bar{\omega}| = \frac{K}{n} |\sum_j \sin(\theta_j - \theta_i)| \leq \frac{K}{n} \cdot n = K$. $\square$

Two-cluster vectors. For $0 < a < n$ let $c^{(a)} \in \mathbb{R}^n$ have its first a entries equal to $p = \frac{n}{2a}$ and the other $n - a$ entries equal to $-q$, $q = \frac{n}{2(n-a)}$ (`clusterFreq n a`). Then $\bar{c}^{(a)} = (ap - (n - a)q)/n = 0$ and $D(c^{(a)}) = (ap + (n - a)q)/n = 1$ (`mean_clusterFreq, meanAbsDev_clusterFreq`). For $n = 6$: $c^{(3)} = (1, 1, 1, -1, -1, -1)$ and $c^{(1)} = (3, -\frac{3}{5}, -\frac{3}{5}, -\frac{3}{5}, -\frac{3}{5}, -\frac{3}{5})$.

Lemma 4 (`clusterPhase_locked, clusterPhase_cohesive`). Let $s_a = \frac{n^2}{2a(n-a)}$ (`clusterThr`). For every $K \geq s_a$, put $\delta = \arcsin(s_a/K) \in [0, \pi/2]$ and $\theta_j = \delta$ for $j \leq a$, $\theta_j = 0$ otherwise. Then $c_i^{(a)} + \frac{K}{n} \sum_j \sin(\theta_j - \theta_i) = 0$ for all i , so $c^{(a)}$ is locked at K . If $K > s_a$, then also $|\theta_i - \theta_j| < \pi/2$ for all i, j .

Proof. $\sin \delta = s_a/K$. For i in the first cluster the sum is $-(n - a)s_a/K$ and $p - (n - a)s_a/n = \frac{n}{2a} - \frac{n}{2a} = 0$. For i in the second cluster the sum is as_a/K and $-q + as_a/n = -\frac{n}{2(n-a)} + \frac{n}{2(n-a)} = 0$. If $K > s_a$ then $s_a/K < 1$ and $\delta < \pi/2$. $\square$

Theorem 5 (`Kc_lt`, via `threshold_separation`). For every $n \geq 6$: $K_c(c^{(3)}) < K_c(c^{(1)})$.

Proof. $s_3 = \frac{n^2}{6(n-3)} < \frac{n}{2}$ because $n^2 < 3n(n-3)$ for $n \geq 5$ (`clusterThr_three_lt`). Let $m = (s_3 + n/2)/2$. By the lemma $c^{(3)}$ is locked at m , so $K_c(c^{(3)}) \leq m < n/2$. The locking set of $c^{(1)}$ is nonempty (it contains $s_1 + 1$) and, by the necessary condition at $i = 1$ with $|c_1^{(1)} - 0| = n/2$, every element is $\geq n/2$; hence $K_c(c^{(1)}) \geq n/2$. $\square$

Corollary 6. For every $n \geq 6$ there is no $F : \mathbb{R} \rightarrow \mathbb{R}$ with $K_c(\omega) = F(D(\omega))$ for all $\omega \in \mathbb{R}^n$ (`Kc_not_function_of_meanAbsDev`). In particular there is no c_1 (even one allowed to depend on n) with $K_c(\omega) = c_1 \sum_i |\omega_i - \bar{\omega}|/n$ for all $\omega \in \mathbb{R}^n$ (`conjecture4082_false`), and no universal c_1 for all n (`conjecture4082_false_universal`).

Proof. $D(c^{(3)}) = D(c^{(1)}) = 1$ would force $K_c(c^{(3)}) = K_c(c^{(1)})$. $\square$

Variant readings. `threshold_separation` is proved for any locking notion P that implies locking at coupling λK ($\lambda > 0$) and holds for $c^{(a)}$ when $\lambda K > s_a$. This gives the same conclusion (no function of D) for: (i) the unnormalised coupling $\omega_i + K \sum_j \sin(\theta_j - \theta_i) = \Omega$ (`conjecture4082_false_unnormalised`, $\lambda = n$); (ii) phase-cohesive locking, where the locked state must also satisfy $|\theta_i - \theta_j| < \pi/2$ (`conjecture4082_false_cohesive`); (iii) the threshold $\inf\{K \geq 0 : \omega \text{ is locked at every } K' \geq K\}$ (`conjecture4082_false_eventual`). Because the family covers every $n \geq 6$, a reading for large n only is refuted as well.

Context. Dorfler and Bullo, *On the Critical Coupling for Kuramoto Oscillators*, arXiv:1011.3878, write: “we present the first explicit necessary and sufficient condition on the critical coupling to achieve synchronization in the finite-dimensional Kuramoto model for an arbitrary distribution of the natural frequencies.” Bronski, Carty and DeVille, arXiv:2007.04343, show “that the set of such frequencies can be expressed by a seminorm which we call the Kuramoto norm.” The proof above does not rely on either paper.

4 Lean formalization and axiom audit

File `Conjecture4082/Basic.lean` (namespace `C4082`, Lean 4.33.1, Mathlib v4.33.1, 346 lines, only `import Mathlib`). Main theorem:

```

theorem conjecture4082_false {n : Nat} (hn : 6 <= n) :
  Not (exists c1 : Real, forall omega : Fin n -> Real,
    Kc omega = c1 * ((sum i, |omega i - mean omega|) / n))

```

lake build succeeds. #print axioms for conjecture4082_false, conjecture4082_false_universal, Kc_not_function_of_meanAbsDev, conjecture4082_false_unnormalised, conjecture4082_false_cohesive, conjecture4082_false_eventual and locked_iff_phaseLockedSolution reports only propext, Classical.choice and Quot.sound. There is no sorry and no native_decide.

Conventions. K_c is a real infimum over $K \geq 0$ (the locking sets used are nonempty and bounded below, so no junk value occurs). Phases are real numbers and the cohesive condition $|\theta_i - \theta_j| < \pi/2$ is imposed on the real representatives. Natural-number indices are cast to $\mathbb{R}$.